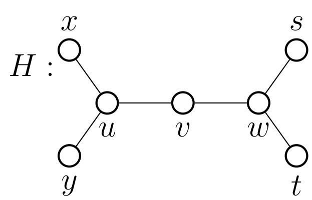
3DM 2024 Proof Workshop 1: Some additional questions

If G is a graph, a path P of G is *maximal* if P is not a proper subgraph of a path. Define

$$\lambda(G) = \max\{\ell(P) : P \text{ is a maximal path of } G\}$$
$$\lambda^{-}(G) = \min\{\ell(P) : P \text{ is a maximal path of } G\}$$

where $\ell(P)$ is the length of P, i.e., the number of edges.

1. Determine $\lambda(G)$ and $\lambda^{-}(G)$ for the graph below



2. Determine λ^{-} and λ for paths and cycles.

3. A graph of the form $P_m \times P_n$ is called a *grid*.

   (a) Determine λ^{-} and λ for $P_2 \times P_5$.

   (b) Determine λ^{-} and λ for $P_2 \times P_n$, where $n \geq 1$.

4. Prove or disprove: $\lambda(G) = \lambda^{-}(G) \iff \Delta(G) \leq 2$.